```
DrFarooqui@raspberrypi:~ $ hostnamectl
 Static hostname: raspberrypi
       Icon name: computer
      Machine ID: fa944534a43c4e73bfd8b54feb33cd82
         Boot ID: 33bb2795a7f84599b6c0b8fb18b845fe
Operating System: Raspbian GNU/Linux 12 (bookworm)
          Kernel: Linux 6.12.34+rpt-rpi-v6
    Architecture: arm
DrFarooqui@raspberrypi:~ $ python miniVNA_Tiny_Calibration_v2.py
```

```
import numpy as np
import json
import socket
from matplotlib import pyplot as plt
local_bt_addr = "00:1A:7D:DA:71:13"  # Your PC's Bluetooth
port = 4

server_sock = socket.socket(socket.AF_BLUETOOTH, socket.SOC
server_sock.bind((local_bt_addr, port))
server_sock.listen(1)

print("Waiting for connection...")
client_sock, client_info = server_sock.accept()
print(f"Accepted connection from {client_info}")

client_sock.settimeout(10)  # 10 seconds timeout

output_filename = "received_data.json"
with open(output_filename, "wb") as f:
    try:
        while True:
            data = client_sock.recv(1024)
            if not data:
                print("No more data received, closing conne
                break
            print(f"Received {len(data)} bytes")
```

```
D:\DrFarooqui\Upwork\James_Clarke\Pyhon_PC_UART\env\python.exe D:\DrFarooqui\Upwork\James_Cl
Waiting for connection...
```

accessing Raspberry Pi Zero W, through SSH from PC

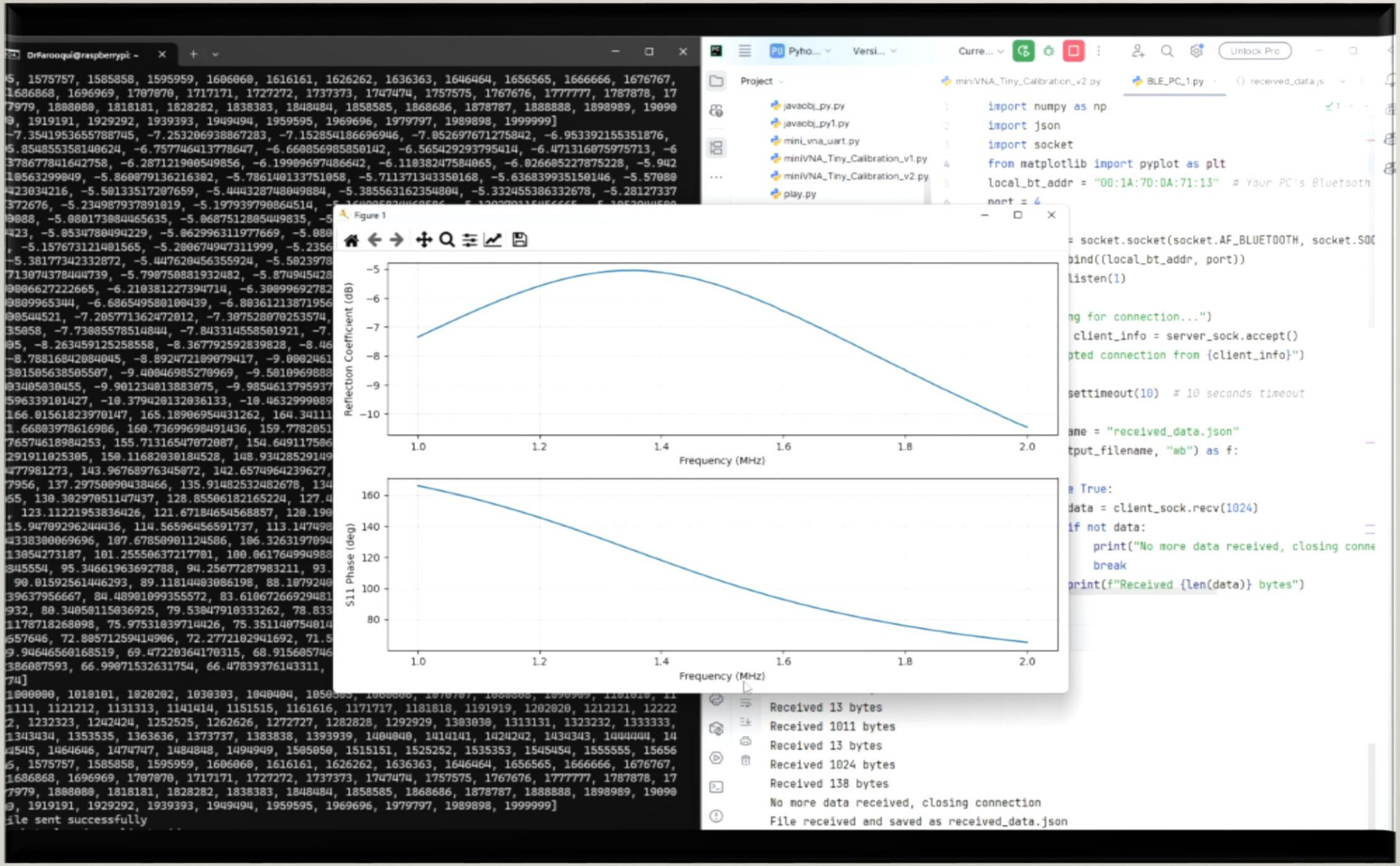
App running on PC to receive data through Bluetooth from Raspberry Pi Zero W

Scenario: Running a Python App on Raspberry Pi Zero W for miniVNA Tiny Calibration

1. Setup Raspberry Pi Zero W:
   - Install Raspberry Pi OS Lite (no GUI/desktop) on the Raspberry Pi Zero W.
   - Install all required Python modules needed to run developed app.

2. Hardware Connections:
   - Power the Raspberry Pi Zero W via USB cable connected to a PC.
   - Connect the miniVNA Tiny to the Raspberry Pi Zero W using a USB cable.

3. Running the Python Application:
   - Execute the Python script miniVNA_Tiny_Calibration_R_Pi_0.py on the Raspberry Pi Zero W.
   - This app performs the following tasks:
     1. Initializes the miniVNA Tiny device.
     2. Loads calibration data previously saved by a Java app (used due to the unavailability of a calibration standard).
     3. Acquires measurement data from the miniVNA.
     4. Applies calibration parameters to the acquired data.
     5. Saves the calibrated data and intermediate debugging data into two separate JSON files.
     6. Sends the JSON file containing calibrated data to the PC via Bluetooth.

4. PC-side Application:
   - On the PC, run the app BLE_PC_App.py which waits to receive data over Bluetooth.
   - Upon receiving the JSON data file, the PC app saves it and plots the S11 magnitude and phase.

Data Acquired